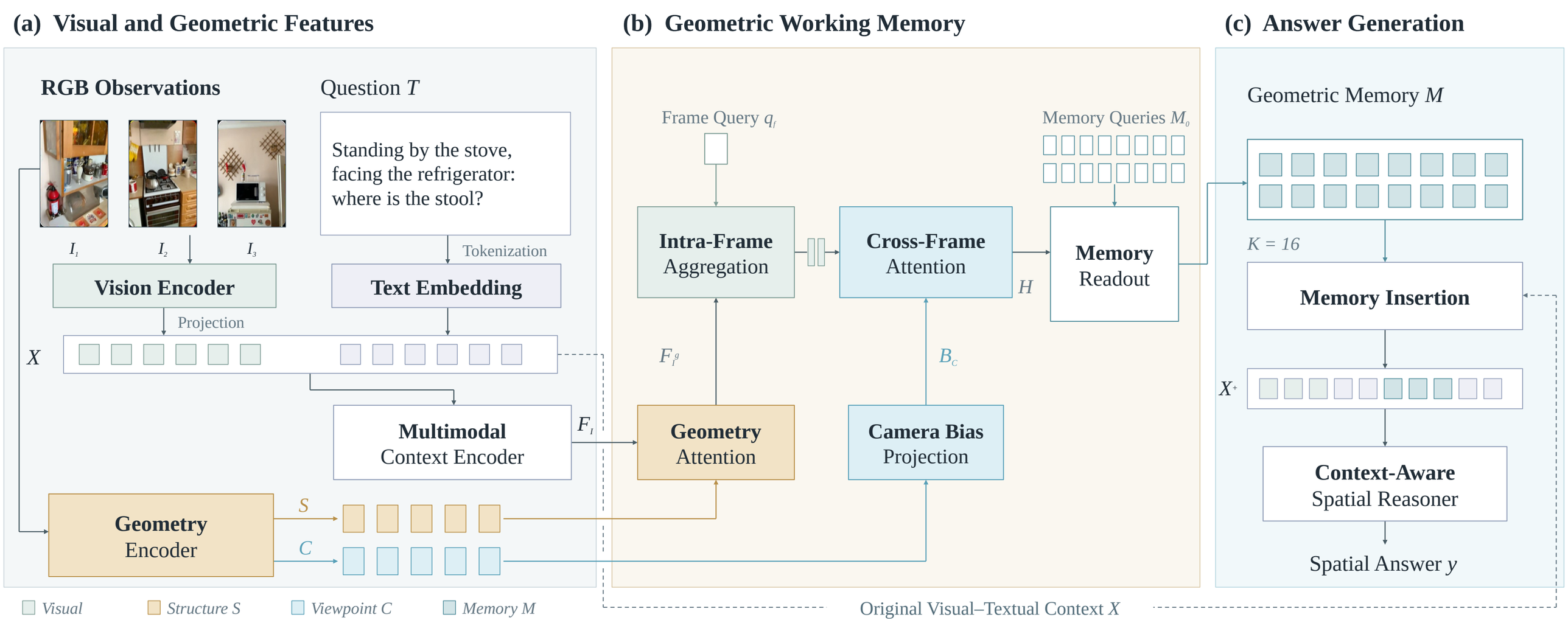

(a) Visual and Geometric Features
(b) Geometric Working Memory
(c) Answer Generation
RGB Observations
Question T
Standing by the stove, facing the refrigerator: where is the stool?
I1
I2
I3
Tokenization
Vision Encoder
Text Embedding
Projection
X
Multimodal Context Encoder
F1
Geometry Encoder
S
C
Frame Query qf
Memory Queries M0
Intra-Frame Aggregation
Cross-Frame Attention
Memory Readout
H
F1g
Bc
Geometry Attention
Camera Bias Projection
Geometric Memory M
K = 16
Memory Insertion
X+
Context-Aware Spatial Reasoner
Spatial Answer y
Visual
Structure S
Viewpoint C
Memory M
Original Visual–Textual Context X